

RNA-seq based sub-typing of breast cancer tissue

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Abstract

Breast cancer is a disease that contains many different molecular distinct subtypes [9]. These subtypes affect the clinical outcomes of patients and are therefore important for treatment selection [11]. The Cancer Genome Atlas is a resource that provides researchers with cancer data, specifically RNA-seq data used in this analysis [12]. Pairing RNA-Seq data with PAM50 subtypes allow for the indication of distinct molecular signatures of subtypes [8]. Differential expression analysis followed by machine learning creates a model that is capable of distinguishing subtypes by RNA counts [2, 3, 6, 7]. Our approach used a consensus feature selection method to determine a 1,734 gene feature set. This feature set resulted in 90% accuracy in subtype classification using a random forest model (n=109). SHapley Additive exPlanations provides explainability of our model, increasing clinical confidence in the results [4]. Notably, all 100 top genes identified through SHAP were also differentially expressed genes identified by PyDESeq2. Our model and approach was validated by recovering experimentally confirmed genes that are specific to certain PAM50 subtypes.

Keywords

Python, Data Mining, scikit-learn, DESeq2, RNA-seq

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1 Introduction

Breast cancer is a disease affecting many people globally, but within the broad "breast cancer" classification, there are many different forms that this disease can take. There are 4 main subtypes of breast cancer: Luminal A, Luminal B, HER2-enriched, and Basal type. Determining subtype clinically relies on immune based methods, but modeling the disease based on gene expression can provide a more comprehensive understanding of the molecular fingerprint of the disease. Determining the subtype of cancer is essential for developing a treatment plan. If the subtype is misclassified, the wrong treatment could be prescribed. Although there are existing methods of developing subtypes, exploring the molecular fingerprint of

these subtypes could reveal sub-groups even within subtypes. Additionally, medicine is a conservative field when it comes to machine learning, so explainability is essential and not just a nice addition. Existing studies have developed models that can classify subtypes of breast cancer but often do not address which genes drive these predictions. Other studies perform differential expression analysis, illuminating statistically distinct genes between subtypes, but without interrogating which gene combinations drive classification predictions. This work developed a reproducible Python pipeline that integrates differential expression analysis, machine learning modeling, and SHapley Additive exPlanations (SHAP) for individual gene influence on classification per sample, enabling trust in the model. The trained model achieved 90% accuracy, partially due to a core contribution of this work: utilizing cross-method convergence of genes by combining multiple feature selection methods to select the strongest candidates for model training. Finally, we validated the pipeline through demonstrating that our approach can recover experimentally validated breast cancer genes, providing a trustworthy pipeline and model for future development or adaptation.

2 Related Work

Previous work described the diversity of breast cancers through clustering them based on their molecular fingerprints, opening the door for the exploration of the disease from a subtype perspective and not as a single disease [9]. Others expanded on this work by further refining the subtype classifications and worked to connect these subtypes to clinical relevance, tying the molecular biology to tangible patient results, confirming the importance of the subtype classification [11]. Later in 2009, a group described PAM50 the 50 gene classifier that is used as a ground truth label in The Cancer Genome Atlas (TCGA). TCGA has collected an enormous amount of data and has become a comprehensive source for data to understand the molecular underpinnings of breast cancer disease [8]. Several authors have used machine learning approaches to analyze data from TCGA. These approaches span from basic random forest models to convolutional neural networks [2, 6]. Recent work uses DeepInsight for explainability and discovery of genes effect on the model training, notably in image data. This is a different approach than the SHAP based machine learning explainability approach. While DeepInsight is interesting, for the purposes of my analysis I chose to use SHAP because it outputs per-sample gene effect scores as opposed to DeepInsight which outputs data as images, which would not be consistent with our approach. Additionally, while machine learning methods have been endlessly applied to TCGA datasets, the work within this proposal extends beyond simply machine learning by using SHAP for explainability and using a multiple method approach for selecting features to ensure that

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genes/biomarker, which serve as features, are selected by a convergence analysis. We will be using standard, well documented tools for the analysis of the data. Differential expression analysis will be done with PyDESeq2, a python port of the original DESeq2 packager for R [3, 7]. SHAP will be used for explainability and interpretation of genes that drive classification and the direction of the effect, allowing for the generation of insights [4]. UMAP can be used for clustering to identify distinct populations of samples [5]. Finally a standard protocol and workflow for working with RNA-seq datasets has been released and can be used as a reference to the work completed in this project [10]. While many of these methods have been used in various combinations on various TCGA datasets, the utility of this study lies in the aim of developing a consensus feature set that can be compared in magnitude and direction using SHAP to generate rankings for target candidates. This pipeline will serve as a resource for similar studies and will be validated by the presence of experimentally validated genes that contribute to the distinction between the different subtypes of breast cancer.

3 Proposed Work

The dataset that I used is from The Cancer Genome Atlas containing breast invasive carcinoma samples with RNA-seq data. All samples have PAM50 labels to describe which subtype of breast cancer that sample is, and a clinical features table that was not used in this analysis but could be used to expand upon these results in a secondary analysis. The analysis used the RNA-seq data which covers 60,660 genes (features) and 546 samples after filtering for only samples which have a PAM50 label. What the RNA-seq data actually is is a quantified amount of RNA from the sample, with the logic being that distinct cell types contain RNA in differing proportions depending on the individual cell type. For this project, I used pandas and numpy in the data wrangling phase. The package mygene was used to convert ensemble IDs of genes to standard gene symbols. For differential expression analysis and normalization PyDESeq2 was used because it is specifically designed to work with the type of genetic data analyzed. I trained models with scikit-learn, a standard package for training models. For explainability shap was used, for its ability to output per-sample effects of each gene with the direction of the effect on classification. Finally, visualizations were completed with seaborn for simple visualizations and matplotlib when more complicated visualizations were needed.

3.1 Data Exploration

Raw RNA (STAR) counts, clinical annotation, and PAM50 subtypes were all pulled from UCSC Xena. TCGA identification codes were truncated to 16 characters so all datasets aligned and could be merged. An inter join was performed across all datasets, yielding a clinical dataset and RNA-seq dataset with PAM50 labels on both, spanning all five subtypes (LumA 232, LumB 129, Basal 98, Her2 58, Normal 30). The STAR counts arrived already transformed with a $\log_2(x+1)$ transformation, so a transformation to undo this was applied to get the raw counts from the RNA-Seq. This is important for PyDESeq2. Genes were filtered down to restrict the dataset to contain only genes which have at least 15 reads across 15 samples. This eliminates many genes that are not used in breast tissue and

therefore would be uninformative. 15 was chosen so that the Normal class $n=30$ would still be preserved. This operation reduced the dataset from 60,660 genes to 27,516. Normalization was applied using variance-stabilizing transformation (VST) via PyDESeq2. PCA, t-SNE, and UMAP were applied to the dataset as dimensionality reduction methods to identify clusters in the data (Figure 1). To much delight with all three methods, the data clustered into what appeared to be 5 clusters. The Basal samples clustered entirely cleanly, with the Normal samples almost completely cleanly. The LumA, LumB, and HER2 samples contained some overlap, but were mostly clustered with members of the same class. A sample-to-sample correlation cluster map of the top 500 most variable genes in the dataset confirmed the block structure by PAM50 subtype. After clustering, PyDESeq2 was fit on the full 5 class dataset with Cook's-distance refitting. 2,265 outlier genes were re-fit. All ten pairwise comparisons were extracted to identify differentially expressed genes (DEGs) between each of the subtypes against one another. For each comparison the list of DEGs were saved with individual lists for full, significant, up-regulated, and down-regulated, relative to the subtype compared against. Volcano plots and heatmaps were generated for each comparison. Volcano plots visualize the up and down regulated genes visually with genes that are not statistically significant in the middle. Down regulated genes are shown in blue with up-regulated in red. Heatmaps allow for the comparison of the most significant DEGs from each comparison to be visualized across all subtypes, showing patterns in their organization.

3.2 Feature Selection

Before training a classifier, feature selection is essential to narrow down the filtered set of 27,516 genes to a feature set that trims down to only essential features while remaining broad enough to accurately capture the differences between subtypes. As stated, a consensus feature selection process is used to achieve this (Figure 2). Three methods are used to generate independent feature sets. First, the union of all DESeq2 genes across all 10 pairwise comparisons were aggregated into a list, 14,037 genes. Then variance across the dataset was computed and the top 10,000 most variable genes were selected with the rationale that these genes were likely to be the most informative since they vary across the populations of subtypes, the second feature set. The last feature set was computed by training a random forest model with the 27,516 gene feature set and using the computed feature importance. The top 5,000 most important genes from training this model were used as the final feature set. After all three feature sets were computed, the intersection of the three was taken to become the consensus feature set that will be used for the classification tasks and SHAP analysis.

3.3 Classification

The data which was split in an 80/20 train/test split, stratified by PAM50 subtype to preserve class distributions across the split, produced a training set ($n=437$) and a test set ($n=109$). The sets were filtered down to only include the features from our consensus feature set. With our training set, I trained 3 classifiers: Random forest, SVM, and Logistic regression. These classifiers were chosen to include both linear and non-linear methods. Hyperparameters for each model were selected by 5-fold stratified cross validation on

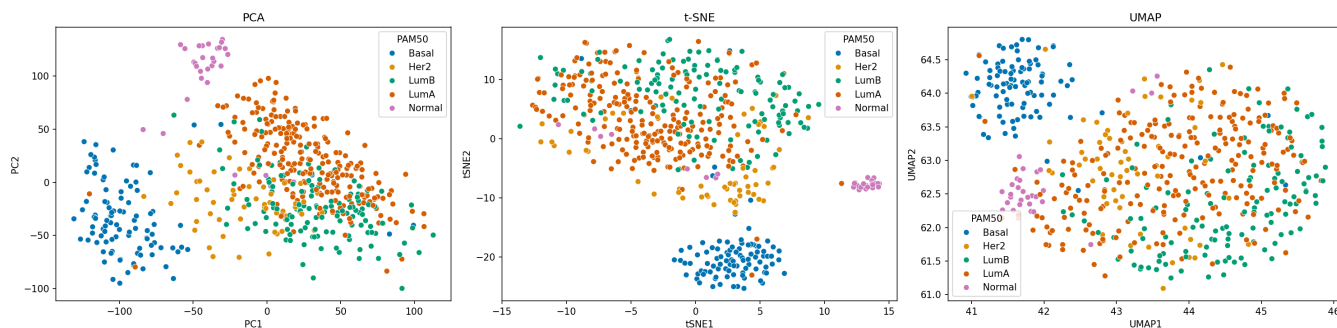


Figure 1: Unsupervised projections of the VST-normalized expression matrix (546 samples, 27,516 genes) colored by PAM50 subtype. PCA captures broad subtype structure with PC1 and PC2 accounting for roughly 12% and 8% of variance respectively; t-SNE (perplexity 30) and UMAP (n_neighbors=70) recover comparable clustering. All three methods cleanly separate Basal and Normal samples, with HER2, LumA, and LumB overlapping but biased toward same-subtype neighbors.

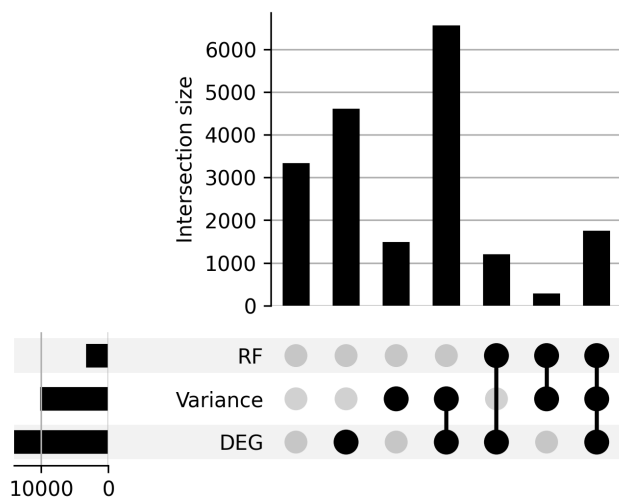


Figure 2: Cross-method overlap of all three feature selection methods, computed from the training set. The three input sets are the union of PyDESeq2 DEGs across all to pairwise PAM50 comparisons (14,037 genes), the top 10,000 genes by training set variance, and the top 5,000 genes by importance from training a random forest model with all features from the training set. The intersection of all three methods produces 1,734 genes, the consensus feature set used for all downstream classification tasks and SHAP analysis.

the training set using GridSearchCV, with macro-averaged F1 as the target metric to use in tuning. Macro F1 was used as opposed to accuracy due to concerns about the class imbalance in the dataset. The selected hyperparameters from optimization are summarized in Table 1.

The top performing models from hyperparameter tuning were then evaluated and analyzed using confusion matrices and ROC curves.

Table 1: Hyperparameter grids and values selected by 5-fold stratified cross-validation on the training set, optimizing macro F1

Model	Grid	Selected
Random Forest	n_estimators: 100, 500, 1000	100
	max_depth: None, 10, 20	None
	min_samples_leaf: 1, 5, 10	1
SVM (RBF)	C: 0.1, 1, 10, 100	10
	gamma: scale, auto	scale
Logistic Reg.	C: 0.001–100 (6 values)	0.1
	l1_ratio: 0.0, 0.5, 1.0	0.0

4 Evaluation

4.1 Differential Expression

All ten pairwise PAM50 comparisons produced large sets of DEGs (Table 2), ranging from 1,711 (LumB vs LumA, the most molecularly similar pair and the most likely source of classifier confusion) to 7,531 (Basal vs LumA, the most distinct pair). Volcano plots for the six non-Normal pairwise contrasts are shown in Figure 3. The Basal-vs-LumA contrast recovered all four experimentally known markers of LumA (ESR1, PGR, FOXA1, GATA3) as significantly down-regulated and the genes associated with Basal cells: KRT5, KRT14, and KRT17, as up-regulated, all at $p_{adj} < 10^{-10}$ (Table 3). The same separation is visible at the sample level in the top-50-DEG heatmap of Basal vs LumA (Figure 4), where samples block cleanly by subtype. Recovery of these established markers without supervision provides confidence that downstream feature selection and SHAP attributions will be biologically interpretable rather than noise-driven.

4.2 Classification Performance

The random forest model achieved 90% accuracy on classifications for the test set. This was the highest of all three models trained (Table 4). This was unexpected that the random forest model performed the best, given that its F1 macro score from hyperparameter tuning was the lowest of all three models. When reviewing the

Pairwise Differential Expression: PAM50 Subtypes

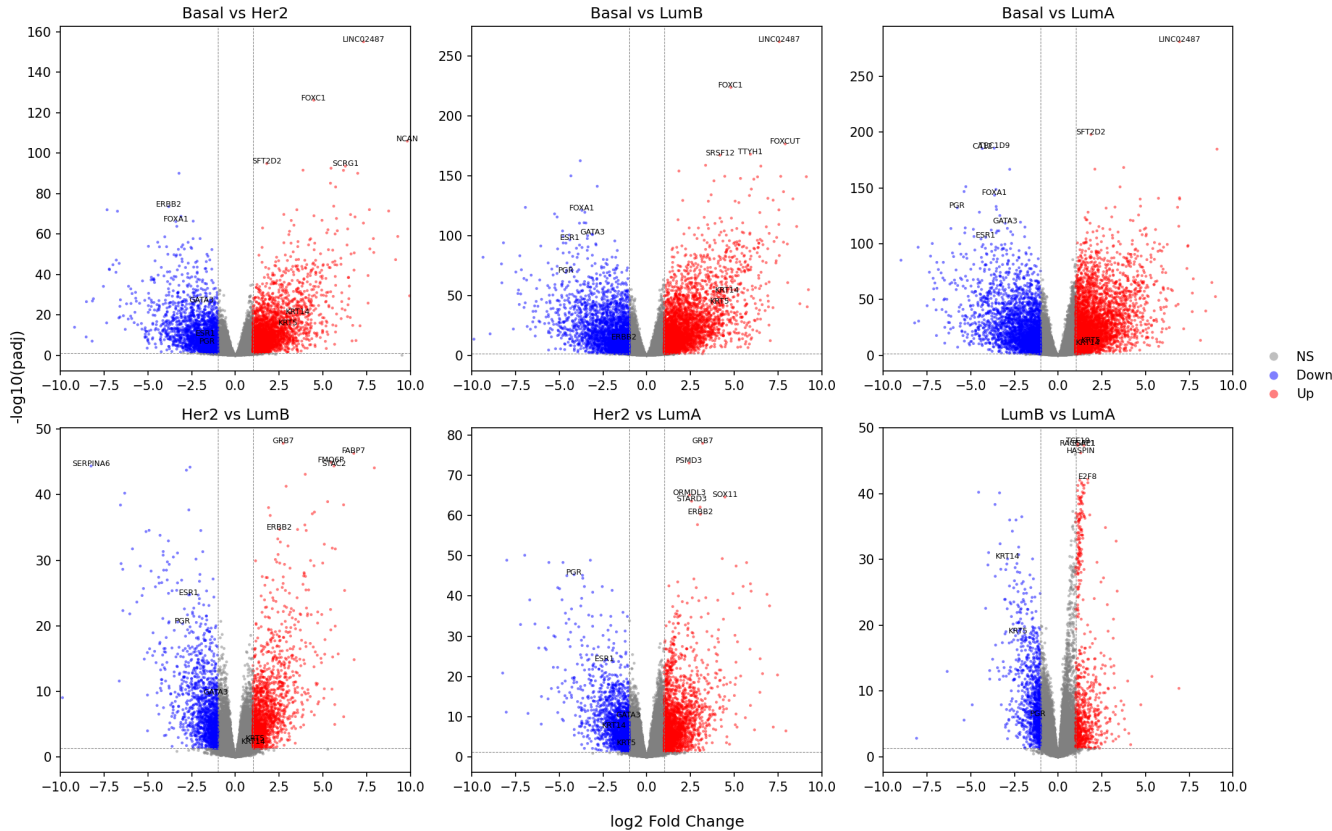


Figure 3: Pairwise differential expression for the six non-Normal PAM50 subtype contrasts. Red points are up-regulated in the first group, blue points are down-regulated, and gray points fail significance thresholds ($p_{adj} < 0.05$, $|\log_2 FC| > 1$). Top five most-significant genes per panel and any significant canonical PAM50 markers are labeled. Per-panel y-axis ranges differ to preserve the relative magnitude of significance within each contrast.

Table 2: Differentially expressed genes (DEGs) per pairwise PAM50 contrast at $p_{adj} < 0.05$ and $|\log_2 FC| > 1$.

Comparison	DEGs	Up	Down
Basal vs Her2	5,418	3,078	2,340
Basal vs LumB	6,966	4,041	2,925
Basal vs LumA	7,531	4,415	3,116
Basal vs Normal	7,521	3,839	3,682
Her2 vs LumB	3,183	1,756	1,427
Her2 vs LumA	3,953	2,113	1,840
Her2 vs Normal	7,420	3,292	4,128
LumB vs LumA	1,711	770	941
LumB vs Normal	7,268	3,076	4,192
LumA vs Normal	5,319	2,111	3,208

Table 3: Recovery of canonical PAM50 markers in the Basal vs LumA contrast. Negative $\log_2 FC$ indicates higher expression in LumA.

Gene	$\log_2 FC$	p_{adj}	Role
FOXA1	-3.63	3.2×10^{-144}	luminal TF
PGR	-5.76	6.8×10^{-133}	progesterone receptor
GATA3	-3.02	4.5×10^{-119}	luminal TF
ESR1	-4.14	1.7×10^{-106}	estrogen receptor
KRT17	+2.10	1.1×10^{-18}	basal cytokeratin
KRT5	+1.88	3.3×10^{-12}	basal cytokeratin
KRT14	+1.71	3.1×10^{-10}	basal cytokeratin
ERBB2	-0.74	1.4×10^{-6}	HER2 receptor

confusion matrix and ROC curve (Figure 5), it becomes clear that the majority of the drop in accuracy is due to confusion between LumA and LumB. This issue was present in all three models. This

makes sense biologically since these two subtypes are very similar on the molecular level as we showed previously with them having the least amount of DEGs ($n=1711$).

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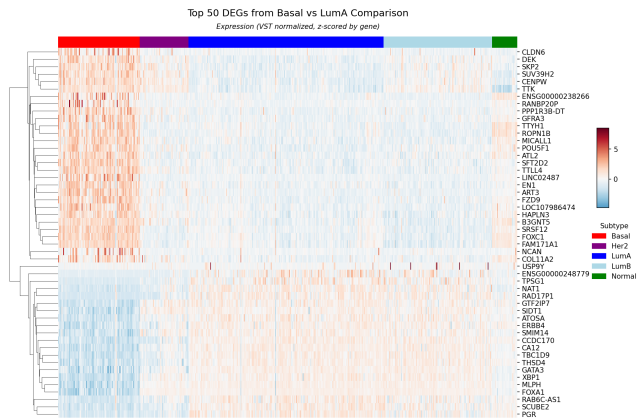


Figure 4: Top 50 differentially expressed genes between Basal and Luminal A samples, VST-normalized and z-scored by gene. The column color bar indicates PAM50 subtype. Samples cluster cleanly into Basal-high and LumA-high blocks; canonical luminal markers (ESR1, PGR, FOXA1, GATA3) and basal cytokeratins (KRT5/14/17) appear at opposite ends of the gene axis.

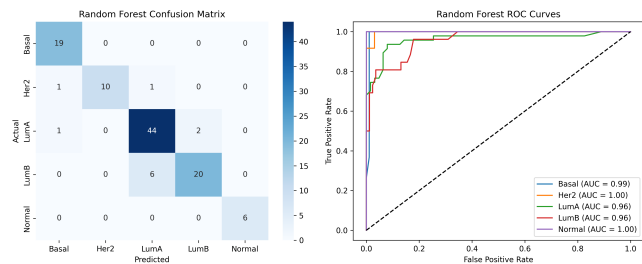


Figure 5: Random Forest classifier performance on the held-out test set ($n = 109$). Left: confusion matrix of predicted versus actual PAM50 subtype. Right: per-class one-vs-rest ROC curves with AUCs reported in the legend. The dominant residual error is LumB samples predicted as LumA, consistent with the small DEG count between those two subtypes (Table 2).

Table 4: Model performance on test set

Model	CV macro F1	Test acc.	Test macro F1
Random Forest	0.828	0.900	0.918
SVM (RBF)	0.855	0.873	0.893
Logistic Regression	0.859	0.882	0.899

4.3 SHAP Interpretation

SHAP analysis produced a ranked list of the most important genes for random forest model. When reviewing the top 100 genes that were important for the classification process, all 100 of them were

DEGs, meaning they were found to be differentially expressed between 2 subtypes with statistical significance. The analysis also revealed many of the experimentally known markers and genes implicated in these subtypes, validating the approach as a whole. We were able to recover these canonical genes purely based on their expression in the samples. For a concrete example, we can investigate the case of ESR1 expression in the basal subtype (Figure 6). Basal-like breast cancer is known to be ESR1 negative, meaning that it should not be expressed at high levels in these cells. This biological fact is illustrated by the results of our SHAP analysis. Lower levels of ESR1 result in a positive SHAP value, meaning that the model is positively influenced toward the Basal class. The higher the level of ESR1, the lower the SHAP value becomes. High levels of ESR1 push the model classification away from the Basal class as shown by the negative SHAP values.

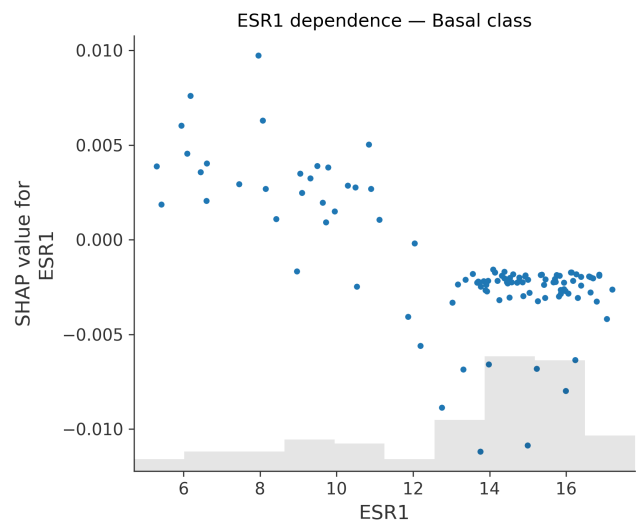


Figure 6: SHAP dependence plot for ESR1 in the Basal class. Each point is one held-out test sample. The x-axis is the VST-normalized ESR1 expression value and the y-axis is the SHAP value contributed by ESR1 to the Basal-class prediction for that sample. Samples with low ESR1 expression have positive SHAP values which pushes the model toward a Basal prediction. Samples with high ESR1 expression have negative SHAP values which pushes the model away from Basal. This is consistent with the established biology that Basal-like breast cancer is estrogen-receptor-negative.

5 Discussion

5.1 Process and Timeline

The project timeline was appropriate for the depth of the project and the analysis completed. While some breaks were taken which caused some delays, overall the project proceeded smoothly. Each major step was a significant milestone with the SHAP analysis taking slightly longer than expected. The writing has required much more time than initially expected, but did not cause significant delay.

5.2 Lessons Learned

- (1) After hyperparameter tuning, I did not expect for the random forest model to perform as well as the others given that it scored a lower macro F1 score. However, after comparison on the test set, it was found that the random forest model performed the best. This was a surprise, and certainly a good lesson to be thorough with analysis, as I could have missed out on this performance if I only tested the best performing model from hyperparameter tuning.
- (2) Another lesson, was that PyDESeq2 was much easier to run once instead of trying to do all pairwise combinations as individual runs. This approach allowed for me to extract the pairwise comparisons instead of having to do repetitive data transformation and analysis.
- (3) The major contribution of this paper, the consensus feature selection, was a major point of learning. Prior to this paper I had not had experience with Upset plots, and I learned how essential proper feature selection is for creating models that perform well to protect from overfitting and preserve enough features to keep the model useful. It is a delicate balance and working with such a large dataset was a good exercise in this balance.

5.3 Limitations

A few limitations certainly exist within this study. One known issue with the analysis is that VST normalization was applied before the train/test split, which can lead to a small amount of data leakage since the entire set was normalized at once instead of allowing for the train and test split to be normalized independently of one another. This was determined to be a small enough issue that it was not addressed, but a more comprehensive analysis could address this gap. The other limitation exists around the LumA and LumB classification. These were the two subtypes that all models did not classify well. While this makes sense, a layered approach, where LumA and LumB are grouped for initial classification could yield better results.

5.4 Future Work

Future studies could expand on this study by addressing the stated limitations. Particularly, addressing the LumA and LumB classification problem could yield a model with incredible accuracy. This could be done by training a model that groups LumA and LumB together into 1 class, Lum. This would be possible due to how biologically similar these classes are. Then, a second classifier for the Lum class could be trained to differentiate between LumA and LumB. This new classifier could be specific to these two subtypes so the influence of the other subtypes can be ignored in this model. The results of this study could be expanded upon by introducing pathway-level interpretation of the results using GSEAPy [1]. This would expand on the SHAP results by matching individual genes into biological processes. This would produce a ranked list of biological processes so the genes could be related to the processes they contribute to, giving biological context to the results.

6 Conclusion

6.1 Summary

Starting from raw RNA-seq data, this work developed a pipeline that produced a ranked list of genes associated with different cancer subtypes. Differential expression analysis revealed genes that are expressed at statistically significant levels between subtypes. This data was combined with other approaches to produce a consensus feature set that produced a random forest model boasting 90% accuracy in subtype classification. Much of the loss of accuracy is attributed to LumA and LumB, biologically similar subtypes, and an approach to address this has been proposed. Additionally, we validated the pipeline through the recovery of genes known to be significant in specific subtypes.

6.2 Key Findings

- (1) A multi-method consensus feature set produced a highly accurate machine learning model
- (2) 90% accuracy was achieved on classification of the test set
- (3) SHAP recovered known markers of the subtypes, validating the pipeline
- (4) 100% of the top-100 SHAP genes were also DEGs

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